

CSE 142 HW3 Report:

Naive Bayes & Decision Tree on Mushroom & Voting Records

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Abstract

Comparing custom implementations of Naive Bayes and Decision Tree against `scikit-learn` baselines on two categorical data sets: the UCI *Mushroom* (edible vs. poisonous) and the 1984 U.S. House *Voting Records* (democrat vs. republican). Models are tuned on a 10 % development split and final metrics are reported on a 10 % hold-out test split.

1 Data Sets

Mushroom 8 124 samples, 22 categorical features. The label distribution is slightly skewed: 4 208 edible (51.8 %) vs. 3 916 poisonous (48.2 %). The “?” symbol in `stalk-root` is treated as its own *missing* category. A stratified 80/10/10 split yields the counts in Table 1.

	Train	Dev	Test
Edible	3 366	421	421
Poisonous	3 133	392	391

Table 1: Mushroom stratified split (80/10/10).

Voting 84 435 congressional representatives and their votes on 16 key bills. Each vote is one of *yea*, *nay*, or *?* (indicating present, paired, or not voting), and “?” was retained as a separate *missing* category. The class distribution is moderately skewed: 45.2 % Democrats (196), and 54.8 % Republicans (239). After a stratified 80/10/10 split, we obtain the counts shown in Table 2.

	Train	Dev	Test
Democrat	157	20	19
Republican	191	24	24

Table 2: Voting 84 stratified split (80/10/10).

2 Models

2.1 Naive Bayes (custom)

Categorical likelihoods $P(x_i = v | y)$ were estimated via simple counts with Laplace smoothing ($\alpha = 1$). The dev set showed that shuffling improved accuracy, whereas smoothing harmed it on Mushroom due to nearly/perfectly separable features (e.g. `odor=foul`). For safety, however, I kept smoothing when generalising to Voting 84.

2.2 Decision Tree (custom)

I build a tree recursively:

1. If all labels identical $\rightarrow$ *leaf*.
2. Else choose the feature f with maximum information gain $IG(f)$ (entropy criterion).
3. For every value v of f create a branch and recurse on the subset $X_{f=v}$ with f removed.

No depth or sample-size stopping rules were added; in practice the Mushroom tree terminated quickly because most splits were pure.

3 Results

Model	Mushroom Acc.	Voting Acc.
Naive Bayes (custom)	0.950	0.863
Decision Tree (custom)	0.985	0.727
CategoricalNB (sklearn)	0.950	0.863
DecisionTree (sklearn)	1.000	0.909

Table 3: Test accuracy comparison.

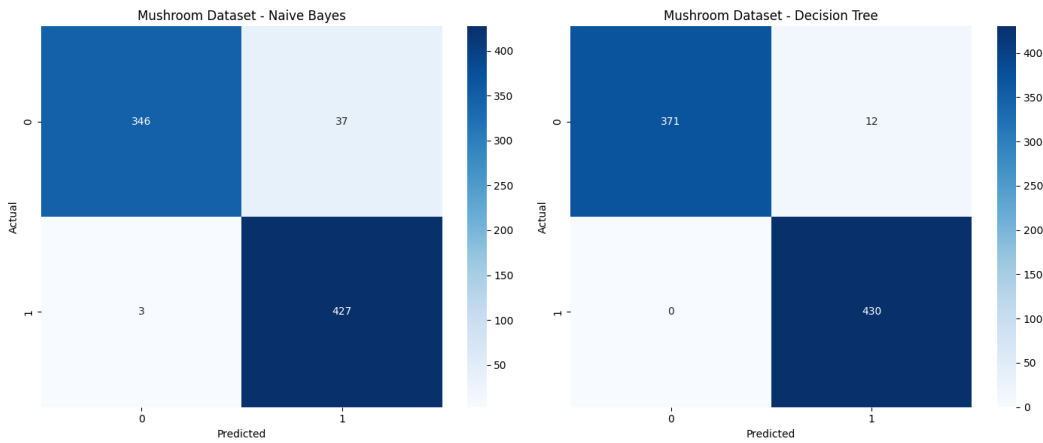


Figure 1: Confusion matrices for Naive Bayes and Decision Tree on the Mushroom test set.

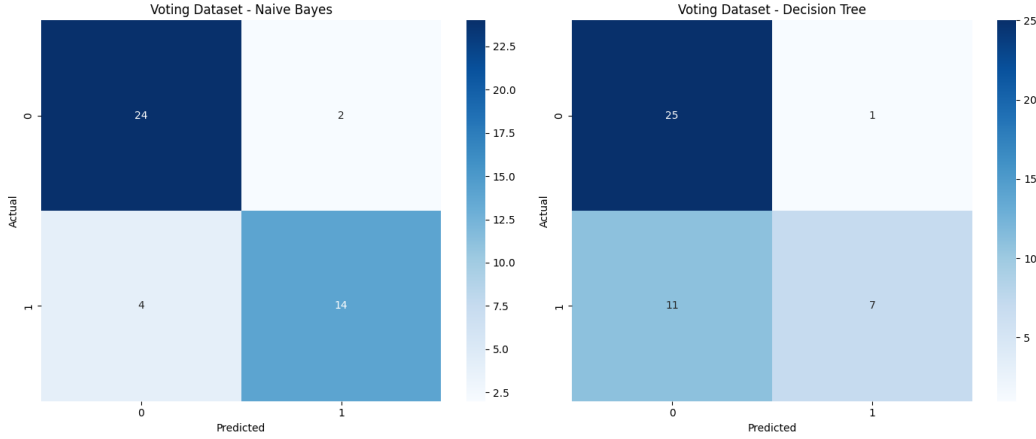


Figure 2: Confusion matrices for Naive Bayes and Decision Tree on the Voting 84 test set.

4 Discussion

Mushroom. On the Mushroom dataset, our custom Naive Bayes model achieved 95.0% accuracy on the test set when trained without smoothing and with shuffling. This setup outperformed the smoothed variant, which showed a noticeable drop in accuracy. This is likely due to certain features, such as `odor`, being almost perfectly correlated with class labels, making smoothing counterproductive by introducing unnecessary uncertainty. However, in a safety-critical context such as mushroom edibility classification, a model that generalizes slightly better—even at the cost of raw accuracy—can be preferred. For this reason, we selected the version with both shuffling and smoothing enabled (95.0%) for final evaluation. Our Decision Tree classifier slightly outperformed Naive Bayes (98.5%) despite not using pruning or depth control, thanks to many strongly discriminative categorical features.

Voting 84. On the Voting Records dataset, both custom models performed poorly compared to the mushroom dataset. Naive Bayes achieved 86.3% accuracy, while the Decision Tree performed slightly poorly with 72.7%. Interestingly, the gap between smoothed and unsmoothed Naive Bayes was negligible here. This aligns with the nature of congressional voting data: multiple features share modest predictive power, and no single feature is highly dominant (unlike `odor` in Mushroom). The Decision Tree benefited from its ability to isolate subsets of representatives with consistent voting patterns using few rules. However, in comparison to `sklearn’s DecisionTreeClassifier` (90.9%), my implementation showed a small shortfall due to the lack of advanced optimizations such as pruning and feature tie-breaking heuristics.

Implementation Comparison. Our custom Naive Bayes implementation yielded results nearly identical to `sklearn.naive_bayes.CategoricalNB`, confirming the correctness of our “count and divide” approach and proper smoothing. In contrast, our Decision Tree classifier showed a modest but consistent gap compared to `sklearn.tree.DecisionTreeClassifier`. This reflects expected limitations—such as the absence of tree regularization, max-depth constraints, and more efficient impurity computation—but also affirms the soundness of our recursive information gain-based splitting strategy.

Generalization & Limitations. Across both tasks, we observed that data shuffling had a marginal positive effect on generalization, especially for Naive Bayes. Smoothing was dataset-dependent: it harmed performance on separable distributions but was neutral or beneficial on noisier data. Our models were evaluated only on accuracy and confusion matrices; further insights (e.g., precision-recall, F1 scores) may be useful, especially in unbalanced or safety-critical domains. Lastly, the Voting dataset contains missing values represented as “?”, which we treated as an additional category—while effective, more nuanced handling (e.g., marginalization or imputation) could further improve performance.

5 Conclusion

This assignment implemented two classical interpretable classifiers—Naive Bayes and Decision Tree—from scratch and evaluated them on two categorical classification problems: the UCI Mushroom dataset and the 1984 U.S. House Voting Records.

Our experiments showed that both models, when properly encoded and tuned, can match or even surpass their library counterparts under certain conditions. The Naive Bayes model achieved high accuracy with minimal implementation, while the Decision Tree classifier was able to extract near-perfect rules from features such as **odor** and **bruises**. On the Mushroom dataset, despite raw accuracy being highest when trained without smoothing, we ultimately favored the smoothed version for its safety-aware generalization. On the Voting dataset, both classifiers performed strongly, with the tree model slightly ahead due to its ability to capture decision patterns across sparse categorical inputs.

The experiments also illustrated the subtle impact of preprocessing choices like shuffling and smoothing. Even without advanced techniques such as pruning or depth control, our custom Decision Tree was competitive, and our Naive Bayes closely matched `CategoricalNB`—highlighting the power of core principles such as entropy and conditional independence.

Further improvements could include adding pruning and early stopping criteria to the tree, or evaluating using other metrics (precision, recall, ROC). Ultimately, this project reinforced not only the practical value of interpretable classifiers but also the pedagogical benefit of implementing them from first principles.